

Spectral Collisions and Boundary Asymptotics for Zig-zag Eulerian Numbers

Exact minimal recurrences, a crossover law, and growing-edge log-concavity
for OEIS A205497

Research report prepared for Vladimir Reshetnikov

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Abstract

Let $z(n, k)$ count linear extensions with k descents of a naturally labeled n -element fence poset. We study the fixed-column sequences $M_{i,k} = z(i + k + 2, k)$ and the regime in which k increases with n . Starting from established order-polynomial and unit-primitive-matrix formulas, we determine the *exact reduced denominator* of every column generating function. A complete classification of shared eigenvalues gives its degree r_k by a totient sum and yields

$$r_k = \frac{40}{27\pi^2}(k+1)^3 + O(k^2 \log(k+2)).$$

Thus the minimal recurrence has asymptotically $80/(9\pi^2)$ of the order of the familiar unreduced recurrence. We next obtain an explicit uniform relative-error bound for the leading spectral approximation. At the transition $n/(k+3/2) - \log n \rightarrow s$, its relative correction tends to $\exp(-e^{-s})$. Finally, for every fixed $0 < c < 1/2$, all sufficiently large rows are strictly log-concave at every index $1 \leq k \leq cn/\log n - 2$, and at the reflected right-edge indices. These are not proofs of full-row log-concavity or real-rootedness. The report separates prior results from the additional theorems, records two OEIS presentation corrections, and supplies exact rational certificates, independent finite enumeration checks, and a further research program.

Status. This is an unrefereed mathematical research manuscript with ordinary proofs and reproducible calculations. It is not a Lean formalization. The repository and OEIS entries were not modified. A targeted literature search is not an exhaustive priority certification.

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1 The problem and the contribution

The zig-zag Eulerian triangle is a particularly suitable OEIS research target: it combines a concrete permutation statistic, rational generating functions, explicit matrix spectra, and a genuine outstanding shape conjecture. Throughout this report, $\log$ is the natural logarithm, and all asymptotic statements concern positive integers tending to infinity unless otherwise stated.

1.1 What was already known

OEIS A205497 records the triangle $z(n, k)$, its older rectangular arrangement $M_{i,k}$, and several statements still labeled conjectures [1]. Those labels must not be read as evidence that all those statements remain unproved. Xin and Zhong's paper [5], especially Section 5, already establishes the fixed-column rational formulas and their Perron growth rates. Their Remark 5.7 explicitly observes cancellation in a displayed denominator. Petersen and Zhuang [6] develop the descent interpretation, order polynomials, gamma-nonnegativity, and a refined theory. Their Conjecture 6.1 includes full-row log-concavity and real-rootedness.

The results attributed to these papers are the starting point, not claims of new discovery. In particular, finding a proof of an old OEIS formula would not, by itself, constitute a new theorem here.

1.2 The additional questions answered here

The main algebraic question is sharper than rationality:

Which factors and multiplicities actually survive after every cancellation, and what is the *smallest* constant-coefficient recurrence for a column?

The main analytic question is sharper than fixed- k asymptotics:

How far toward the interior may k grow while spectral asymptotics remain uniform, and can those estimates establish a growing part of the log-concavity conjecture?

The answers are summarized below. Here

$$H_m = (\mathbf{1}_{\{i+j \geq m+1\}})_{1 \leq i, j \leq m}, \quad D_m(x) = \det(I - xH_m), \quad D_0 = 1.$$

All polynomial greatest common divisors and least common multiples are normalized to have constant coefficient 1.

Theorem 1.1 (Main algebraic result). *For every $k \geq 0$, the reduced denominator of $C_k(x) = \sum_{i \geq 0} M_{i,k} x^i$ is exactly*

$$R_k(x) = \text{lcm}_{1 \leq m \leq k+1} D_m(x)^{k+2-m}. \quad (1)$$

Its degree r_k is the minimum possible order of a homogeneous constant-coefficient recurrence, even when the recurrence is required only eventually. An explicit totient formula is given in theorem 5.4; in particular,

$$r_k = \frac{40}{27\pi^2} (k+1)^3 + O((k+1)^2 \log(k+2)). \quad (2)$$

For $k \geq 1$, the reduced numerator has degree $r_k - k - 3$.

Theorem 1.2 (Main analytic results). *Put*

$$t_m = \frac{\pi}{4m+2}, \quad \rho_m = \frac{1}{2 \sin t_m}, \quad \alpha_m = \frac{4\rho_m \cos^2 t_m}{2m+1}. \quad (3)$$

Then:

(i) *For every fixed $0 < c < 1$, $z(n, k) \sim \alpha_{k+1} \rho_{k+1}^n$ uniformly over $0 \leq k$ with $k+1 \leq cn/\log n$.*

(ii) *If $n/(k+3/2) - \log n \rightarrow s \in \mathbb{R}$, then*

$$\frac{z(n, k)}{\alpha_{k+1} \rho_{k+1}^n} \rightarrow \exp(-e^{-s}). \quad (4)$$

(iii) *For every fixed $0 < c < 1/2$, all sufficiently large n satisfy*

$$z(n, k)^2 > z(n, k-1)z(n, k+1) \quad \text{whenever} \quad 1 \leq k, \quad k+2 \leq \frac{cn}{\log n}. \quad (5)$$

The same holds at the reflected indices $n-2-k$.

The claims presented as additional contributions are the exact minimality and collision classification, the resulting recurrence-order asymptotic, and the uniform, crossover, and growing-edge theorems. The proofs below establish these statements within this manuscript. Their global bibliographic novelty has not been independently certified.

1.3 Relation to ProveIt

The requested repository is a natural destination for a proof-and-certificate package of this kind. Its README distinguishes exploratory computations from formal theorem boundaries [7]. We follow that distinction: the proofs below do not assume any unverified research claim from the repository, and the Python certificates do not masquerade as kernel-checked proofs. The contribution is standalone and ready for review or later formalization, not a claim that the repository already contains these results.

[Write note, batch 7302, 1 October 2026.] This manuscript is placed in ProveIt as the report `a205497-zigzag-eulerian-spectra` (Section C). The repository contains no other work on A205497 or on fence linear extensions. Its volume *Combinatorial Transseries and Their Inverses* [8] treats two related but different objects: the Euler zigzag numbers A000111 (its chapter `ct:ch:zigzag` on alternating permutations), which are the row sums $\sum_k z(n, k)$ of A205497, and the columns of the classical Eulerian triangle A008292 (its proposition `t1:prop:eulerian`). A205497 is the descent refinement of the linear extensions of the fence, not either of those; nothing here depends on, or is implied by, that volume.

2 Definitions and the order-polynomial bridge

Let P_n be the fence poset with underlying vertices $v_1, \dots, v_n$ and relations

$$v_1 < v_2 > v_3 < v_4 > \dots.$$

Choose any natural labeling: a bijection to $[n]$ that respects every poset comparison. For a linear extension, written as a permutation of its labels, let `des` be its number of adjacent descents. Define

$$A_n(t) = \sum_k z(n, k) t^k.$$

For $n = 0$ set $A_0(t) = 1$. This convention also gives $A_1(t) = A_2(t) = 1$. For $n \geq 2$, the degree is $n - 2$, and the established symmetry is

$$z(n, k) = z(n, n - 2 - k). \quad (6)$$

We use this known symmetry only to transfer left-edge statements to the right edge. All left-edge analysis is independent of it.

Definition 2.1. For $m \geq 1$, let $\Omega_n(m)$ count words $a_1 \cdots a_n$ over $[m]$ with

$$a_1 \leq a_2 \geq a_3 \leq a_4 \geq \cdots.$$

Set $\Omega_0(m) = 1$. These are precisely the order-preserving maps $P_n \rightarrow [m]$. (The array of these counts is OEIS A050446 [2], up to the indexing conventions reconciled in [6, 5].)

Proposition 2.2 (Order-polynomial identity). *For every $n \geq 0$,*

$$\frac{tA_n(t)}{(1-t)^{n+1}} = \sum_{m \geq 1} \Omega_n(m) t^m. \quad (7)$$

Consequently, for every $k \geq 0$,

$$z(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} \Omega_n(k+1-j), \quad (8)$$

where $\binom{n+1}{j} = 0$ for $j > n+1$.

Proof. Sort the vertices by the value of an order-preserving map, breaking equalities by increasing natural label. The result is a unique linear extension. Along a fixed extension with d descents, the values form a weakly increasing sequence in $[m]$, with a compulsory strict increase at each descent. Subtracting the number of preceding compulsory increases gives $\binom{m+n-1-d}{n}$ possibilities, interpreted as zero when $m \leq d$. The generating function of these numbers over $m \geq 1$ is $t^{d+1}/(1-t)^{n+1}$. Summing over extensions proves (7). The case $n = 0$ follows from the geometric series. Multiplication by $(1-t)^{n+1}$ and coefficient extraction prove (8). $\square$

This is the standard P -partition mechanism; its use for this triangle and the equivalent interpretation by big returns of alternating permutations are established in [6]. A big return is a pair of consecutive values $i, i+1$ for which the position of i exceeds that of $i+1$ by more than 1. That interpretation will supply an independent finite check.

n	$(z(n, 0), z(n, 1), \dots)$
2	1
3	1, 1
4	1, 3, 1
5	1, 7, 7, 1
6	1, 14, 31, 14, 1
7	1, 26, 109, 109, 26, 1
8	1, 46, 334, 623, 334, 46, 1

The indexing distinction is essential:

$$M_{i,k} = z(i+k+2, k), \quad C_k(x) = \sum_{i \geq 0} z(i+k+2, k) x^i. \quad (9)$$

If $B_k(x) = \sum_{n \geq 0} z(n, k)x^n$, then

$$C_k(x) = x^{-k-2}B_k(x) \quad (k \geq 1), \quad C_0(x) = B_0(x) = \frac{1}{1-x}. \quad (10)$$

For $k \geq 1$, all the apparently negative powers in the first expression vanish. The separate $k = 0$ formula prevents a spurious boundary error.

3 A complete spectral expansion

The spectrum of the unit-primitive matrix is known; see [5]. We give a proof including the endpoint weights needed for uniform estimates and for excluding pole cancellation.

Lemma 3.1 (The walk model). *For $n \geq 1$,*

$$\Omega_n(m) = \mathbf{1}^T H_m^{n-1} \mathbf{1}. \quad (11)$$

Moreover, H_m is invertible, $H_m^{-1} \mathbf{1} = e_m$, and $(H_m^2)_{ij} = \min(i, j)$.

Proof. Reflect each odd-position letter of a weak alternating word: $b_i = m+1 - a_i$ for odd i , and $b_i = a_i$ for even i . Every adjacent inequality becomes $b_i + b_{i+1} \geq m+1$. Thus the transformed word is a walk counted by (11). Let L be the lower-triangular all-ones matrix and J the reversal matrix. Then $H_m = LJ$, so $\det H_m = \det J = \pm 1$. The last column of H_m is $\mathbf{1}$, giving $H_m^{-1} \mathbf{1} = e_m$. Finally, the (i, j) entry of H_m^2 counts the indices $r \geq \max(m+1-i, m+1-j)$, of which there are $\min(i, j)$. $\square$

Theorem 3.2 (Spectrum with nonzero weights). *For $1 \leq r \leq m$, put*

$$\theta_{m,r} = \frac{(2r-1)\pi}{2m+1}, \quad \lambda_{m,r} = \frac{(-1)^{r+1}}{2 \sin(\theta_{m,r}/2)}, \quad c_{m,r} = \frac{4\lambda_{m,r} \cos^2(\theta_{m,r}/2)}{2m+1}.$$

The $\lambda_{m,r}$ are the distinct eigenvalues of H_m and are arranged in strictly decreasing absolute value. Every $c_{m,r}$ is nonzero, and

$$\Omega_n(m) = \sum_{r=1}^m c_{m,r} \lambda_{m,r}^n \quad (n \geq 0). \quad (12)$$

Furthermore,

$$\sum_r c_{m,r} = 1, \quad \sum_r \frac{c_{m,r}}{\lambda_{m,r}} = 1. \quad (13)$$

Proof. The inverse of $(\min(i, j))_{i,j=1}^m$ is the tridiagonal matrix with -1 on the neighboring off-diagonals, diagonal entries 2 except for the last entry 1. The vectors

$$v_r(i) = \sin(i\theta_{m,r})$$

are its eigenvectors: the usual second-difference identity applies with boundary conditions $v_r(0) = 0$ and $v_r(m+1) = v_r(m)$. Its eigenvalues are $4 \sin^2(\theta_{m,r}/2)$ and are distinct. Since H_m commutes with its inverse square, it preserves each of these one-dimensional eigenspaces. To determine the sign of its eigenvalue, use

$$\sum_{i=1}^m \sin(i\theta_{m,r}) = \frac{1}{2} \cot(\theta_{m,r}/2), \quad v_r(m) = (-1)^{r-1} \cos(\theta_{m,r}/2).$$

The quotient is exactly $\lambda_{m,r}$. Also,

$$\|v_r\|^2 = \sum_{i=1}^m \sin^2(i\theta_{m,r}) = \frac{2m+1}{4}.$$

For example, the last identity follows by summing the nontrivial $(2m+1)$ st roots of unity in symmetric pairs. Orthogonal spectral decomposition of (11) now gives

$$\mathbf{1}^T H_m^{n-1} \mathbf{1} = \sum_r \frac{(\mathbf{1}^T v_r)^2}{\|v_r\|^2} \lambda_{m,r}^{n-1},$$

whose coefficient after extracting $\lambda_{m,r}^n$ is $c_{m,r}$. Invertibility permits the same expression at $n=0$, because $\mathbf{1}^T H_m^{-1} \mathbf{1} = \mathbf{1}^T e_m = 1$. Likewise, $\mathbf{1}^T H_m^{-2} \mathbf{1} = \|H_m^{-1} \mathbf{1}\|^2 = 1$, proving the second identity in (13). Nonvanishing and the ordering of absolute values follow directly from the trigonometric expressions. $\square$

The Perron eigenvalue and its coefficient are $\lambda_{m,1} = \rho_m$ and $c_{m,1} = \alpha_m$ as in (3). For $F_m(x) = \sum_{n \geq 0} \Omega_n(m) x^n$, the theorem gives

$$F_m(x) = \sum_{r=1}^m \frac{c_{m,r}}{1 - \lambda_{m,r} x}. \quad (14)$$

Every pole is present, so its reduced denominator is D_m . The numerator has degree $m-1$, and (13) gives

$$F_m(x) = -x^{-1} + O(x^{-2}) \quad (x \rightarrow \infty). \quad (15)$$

For efficient exact computation we also use the established identity $F_m(x) = 1/(F_{m-1}(-x) - x)$, $m \geq 2$ [6, Theorem 3.15]. Combined with the pole count above, induction yields

$$D_0 = 1, \quad D_1 = 1 - x, \quad D_m(x) = D_{m-2}(x) - x D_{m-1}(-x), \quad F_m(x) = \frac{D_{m-1}(-x)}{D_m(x)}. \quad (16)$$

There can be no canceled factor in this induction, since the reduced denominator already has degree m .

4 Exact column formulas and minimality

Combining (8) and (12) gives an exact, finite expression rather than only an asymptotic expansion.

Proposition 4.1 (Exponential-polynomial expression). *For $n, k \geq 0$,*

$$z(n, k) = \sum_{m=1}^{k+1} \sum_{r=1}^m (-1)^{k+1-m} c_{m,r} \binom{n+1}{k+1-m} \lambda_{m,r}^n. \quad (17)$$

This formula is a consequence of the established spectral and extraction formulas. Its useful feature for the new minimality question is that a shared eigenvalue may occur at several sizes, but the polynomial multipliers then have *different degrees*.

For $j \geq 0$ define

$$\Psi_0(u) = \frac{1}{1-u}, \quad \Psi_j(u) = \frac{u^{j-1}}{(1-u)^{j+1}} \quad (j \geq 1). \quad (18)$$

The binomial-series identity $\sum_{n \geq 0} \binom{n+1}{j} u^n = \Psi_j(u)$ gives

$$B_k(x) = \sum_{m=1}^{k+1} \sum_{r=1}^m (-1)^{k+1-m} c_{m,r} \Psi_{k+1-m}(\lambda_{m,r} x). \quad (19)$$

Alternatively, setting $\vartheta = x d/dx$,

$$B_k(x) = \sum_{j=0}^k (-1)^j \binom{\vartheta+1}{j} F_{k+1-j}(x). \quad (20)$$

The operator binomial is a polynomial in ϑ , divided by $j!$. Both identities hold as formal rational identities.

Proof of the exact-denominator assertion in theorem 1.1. Fix an eigenvalue λ occurring among sizes $1, \dots, k+1$. In (19), its occurrence at size m has a pole of order $j+1 = k+2-m$ at $x = 1/\lambda$. The numerator of Ψ_j is 1 at that point, and the coefficient $c_{m,r}$ is nonzero. Within one size, all eigenvalues are distinct. Among all sizes sharing λ , the smallest size m_0 therefore contributes a *unique* highest-order pole. Lower-order terms cannot cancel it. Its order is exactly $k+2-m_0$, the maximum of the exponents assigned to that root by the factors in (1). This proves the denominator formula for B_k . For $k \geq 1$, multiplication by x^{-k-2} neither removes nor changes any nonzero pole, and the origin is removable by (10). The $k=0$ case is direct. $\square$

Lemma 4.2 (Denominators and recurrence order). *Let $C(x) = N(x)/R(x)$ with $N, R \in \mathbb{Q}[x]$, $R(0) = 1$, $\gcd(N, R) = 1$, and $\deg N < \deg R = r$. Then the coefficient sequence satisfies a recurrence of order r . No recurrence of smaller order can hold for all sufficiently large indices, even with complex coefficients.*

Proof. Write $R = 1 + r_1x + \dots + r_rx^r$. Comparing coefficients of $RC = N$ gives $a_i + r_1a_{i-1} + \dots + r_ra_{i-r} = 0$ for $i \geq r$. If a polynomial Q encoding an eventual recurrence had smaller degree, then QC would be a polynomial, since only finitely many recurrence residuals would be nonzero. Thus R would divide Q after reduction, a contradiction. Equivalently, every pole of C and its full multiplicity must be canceled by Q . This latter argument also covers complex recurrence coefficients. $\square$

Proposition 4.3 (Numerator degree). *Let*

$$\mathcal{P}_k(x) = \prod_{m=1}^{k+1} D_m(x)^{k+2-m}.$$

For $k \geq 1$, writing $C_k = N_k/R_k = G_k/\mathcal{P}_k$, one has

$$\deg N_k = r_k - k - 3, \quad (21)$$

$$\deg G_k = \binom{k+3}{3} - k - 3 = \frac{(k-1)(k+3)(k+4)}{6}. \quad (22)$$

Moreover, $C_k(x) = -x^{-k-3} + O(x^{-k-4})$ at infinity.

Proof. In (19), all terms with $j \geq 1$ are $O(x^{-2})$ at infinity. The $j=0$ terms form $F_{k+1}(x)$, which is $-x^{-1} + O(x^{-2})$ by (15). Use (10). The asserted degrees follow from the nonzero leading Laurent coefficient and $\deg \mathcal{P}_k = \sum_{m=1}^{k+1} m(k+2-m) = \binom{k+3}{3}$. $\square$

The unreduced degree formula is already in [5]; its reduced counterpart now follows from exact minimality.

5 All spectral collisions and the order formula

5.1 A greatest-common-divisor theorem

The reciprocal eigenvalues have the especially useful description

$$\{x : D_m(x) = 0\} = \left\{ (-1)^m 2 \cos \frac{2\pi j}{2m+1} : 1 \leq j \leq m \right\}. \quad (23)$$

To see this directly, write $\ell = m - r + 1$ in the reciprocal of $\lambda_{m,r}$. It becomes

$$(-1)^m 2 (-1)^\ell \cos \frac{\ell\pi}{2m+1}.$$

For even ℓ use $j = \ell/2$; for odd ℓ use $j = (2m+1-\ell)/2$. These indices exhaust $1, \dots, m$.

Theorem 5.1 (Complete collision classification). *If $m, n \geq 1$ have opposite parity, then $\gcd(D_m, D_n) = 1$. If they have the same parity, put*

$$g = \gcd(2m+1, 2n+1), \quad r = (g-1)/2.$$

Then

$$\gcd(D_m(x), D_n(x)) = D_r((-1)^{m-r}x). \quad (24)$$

Proof. All angles in (23) lie strictly between 0 and π . For the same sign, equality of cosines is therefore equality of angles. Common fractions $j/(2m+1) = \ell/(2n+1)$ are exactly a/g , $1 \leq a \leq (g-1)/2$. Their roots have sign $(-1)^m$, which is the root set of the right side of (24). All roots are simple and both polynomials have constant term 1. For opposite signs, equality would instead require the two angles to sum to π . That gives $2(j(2n+1) + \ell(2m+1)) = (2m+1)(2n+1)$, an equality between an even and an odd integer. This is impossible. $\square$

5.2 Primitive factors and their first appearances

For an odd integer $d \geq 3$, let

$$\mathcal{C}_d(y) = \prod_{\substack{1 \leq a \leq (d-1)/2 \\ \gcd(a,d)=1}} \left(y - 2 \cos \frac{2\pi a}{d} \right).$$

This is the real cyclotomic polynomial, of degree $\varphi(d)/2$. For $\varepsilon \in \{1, -1\}$ define

$$Q_{d,\varepsilon}(x) = \frac{\mathcal{C}_d(\varepsilon x)}{\mathcal{C}_d(0)}, \quad \sigma_d = (-1)^{(d-1)/2}. \quad (25)$$

The factors are normalized to have constant term 1.

Lemma 5.2. *The $Q_{d,\varepsilon}$ lie in $\mathbb{Z}[x]$, are irreducible over $\mathbb{Q}$, and are pairwise coprime as (d, ε) varies. The first occurrence of Q_{d,σ_d} among the D_m is at $m = (d-1)/2$; the first occurrence of $Q_{d,-\sigma_d}$ is at $m = (3d-1)/2$.*

Proof. For a primitive d th root ζ , the conjugates of $\zeta + \zeta^{-1}$ are the displayed real numbers, with precisely the identifications $a \sim -a$. The real subfield has degree $\varphi(d)/2$, so $\mathcal{C}_d$ is its monic irreducible polynomial in $\mathbb{Z}[y]$. Its constant term is ± 1 : the algebraic integer $\zeta + \zeta^{-1} = \zeta^{-1}(1 + \zeta^2)$ is a unit because $\Phi_d(-1) = 1$ for odd $d > 1$. For completeness, this last identity follows from $\Phi_d(-x) = \Phi_{2d}(x)$ and the elementary evaluation $\Phi_N(1) = 1$ when $N > 1$ is not a prime

power; that evaluation follows inductively from $N = \prod_{e|N, e>1} \Phi_e(1)$. Thus normalization in (25) preserves integral coefficients and irreducibility.

Distinct primitive odd denominators cannot give the same positive-sign cosine: the corresponding reduced fractions would coincide. Opposite signs cannot coincide by the parity argument in theorem 5.1. To locate a factor in D_m , write $2m+1 = \ell d$ with ℓ odd. Then

$$m - \frac{d-1}{2} = \frac{(\ell-1)d}{2}.$$

This difference is even for $\ell \equiv 1 \pmod{4}$, odd for $\ell \equiv 3 \pmod{4}$. The first choices are respectively $\ell = 1$ and $\ell = 3$, proving the claim. $\square$

Theorem 5.3 (Factorization of the minimal denominator). *Put $M = k + 1$. Then*

$$\begin{aligned} R_k(x) = & \prod_{\substack{d \geq 3 \text{ odd} \\ d \leq 2M+1}} Q_{d, \sigma_d}(x)^{M+1-(d-1)/2} \\ & \times \prod_{\substack{d \geq 3 \text{ odd} \\ 3d \leq 2M+1}} Q_{d, -\sigma_d}(x)^{M+1-(3d-1)/2}. \end{aligned} \quad (26)$$

Proof. The exponent of a factor in the least common multiple is the exponent at its first occurrence, because $M+1-m$ strictly decreases with m . Apply theorem 5.2 to (1). $\square$

Theorem 5.4 (Exact recurrence order). *With $M = k + 1$,*

$$\begin{aligned} r_k = & \frac{1}{2} \sum_{\substack{3 \leq d \leq 2M+1 \\ d \text{ odd}}} \varphi(d) \left(M + 1 - \frac{d-1}{2} \right) \\ & + \frac{1}{2} \sum_{\substack{3 \leq d \leq (2M+1)/3 \\ d \text{ odd}}} \varphi(d) \left(M + 1 - \frac{3d-1}{2} \right). \end{aligned} \quad (27)$$

Equivalently, the number b_m of new reciprocal eigenvalues first appearing at size m is

$$b_m = \frac{\varphi(2m+1)}{2} + \mathbf{1}_{\{3|2m+1, 2m+1 \geq 9\}} \frac{\varphi((2m+1)/3)}{2}, \quad (28)$$

and

$$r_k = \sum_{m=1}^{k+1} (k+2-m)b_m. \quad (29)$$

Proof. Take degrees in (26). The equivalent formulation groups primitive factors by their first appearance; every new root contributes a pole of order $k+2-m$. $\square$

5.3 The cubic asymptotic and compression ratio

Lemma 5.5 (Odd totients). *As $X \rightarrow \infty$,*

$$S(X) := \sum_{\substack{1 \leq d \leq X \\ d \text{ odd}}} \varphi(d) = \frac{2}{\pi^2} X^2 + O(X \log(X+2)).$$

Proof. Using $\varphi(n) = \sum_{d|n} \mu(d)n/d$ and restricting to odd integers,

$$S(X) = \sum_{\substack{d \leq X \\ d \text{ odd}}} \mu(d) \sum_{\substack{r \leq X/d \\ r \text{ odd}}} r.$$

The inner sum is $X^2/(4d^2) + O(X/d)$. The sum of error bounds is $O(X \log(X+2))$, and extending the absolutely convergent main sum to infinity costs $O(X)$. Finally,

$$\sum_{d \text{ odd}} \frac{\mu(d)}{d^2} = \prod_{p \neq 2} (1 - p^{-2}) = \frac{8}{\pi^2},$$

which gives the result. $\square$

Proof of (2). A summation of $S(t)$ gives

$$\sum_{\substack{d \leq X \\ d \text{ odd}}} \varphi(d)(X-d) = \int_0^X S(t) dt = \frac{2X^3}{3\pi^2} + O(X^2 \log(X+2)).$$

In the first sum in (27), set $X = 2M + 3$; the multiplier becomes $(X-d)/4$. Adding a possible zero-weight endpoint is harmless, and deleting $d = 1$ costs only $O(M)$. The main term is therefore $4M^3/(3\pi^2)$. For the second sum set $X = (2M + 3)/3$; its multiplier is $3(X-d)/4$, giving $4M^3/(27\pi^2)$. Their sum is $40M^3/(27\pi^2)$ with the asserted error. $\square$

Since the unreduced denominator has degree $\binom{k+3}{3}$,

$$\frac{r_k}{\binom{k+3}{3}} \longrightarrow \frac{80}{9\pi^2} = 0.9006327434\dots \quad (30)$$

This is an asymptotic reduction of approximately 9.94% in the minimum recurrence order, not a claim about the runtime of a particular implementation.

Table 1: Exact orders and degrees. “Excess” is the degree canceled from the unreduced denominator. The $k = 0$ numerator is treated separately.

k	$\deg \mathcal{P}_k$	r_k	Excess	$\deg N_k$
0	1	1	0	0
1	4	4	0	0
2	10	10	0	5
3	20	20	0	14
4	35	35	0	28
5	56	56	0	48
6	84	83	1	74
7	120	118	2	108
8	165	162	3	151
9	220	215	5	203
10	286	279	7	266
11	364	353	11	339
12	455	439	16	424
13	560	539	21	523
14	680	654	26	637
15	816	784	32	766

6 Consequences and corrections for the OEIS entries

This section concerns presentation and indexing in the entries retrieved on 1 October 2026; it does not claim the underlying fixed-column formulas as new.

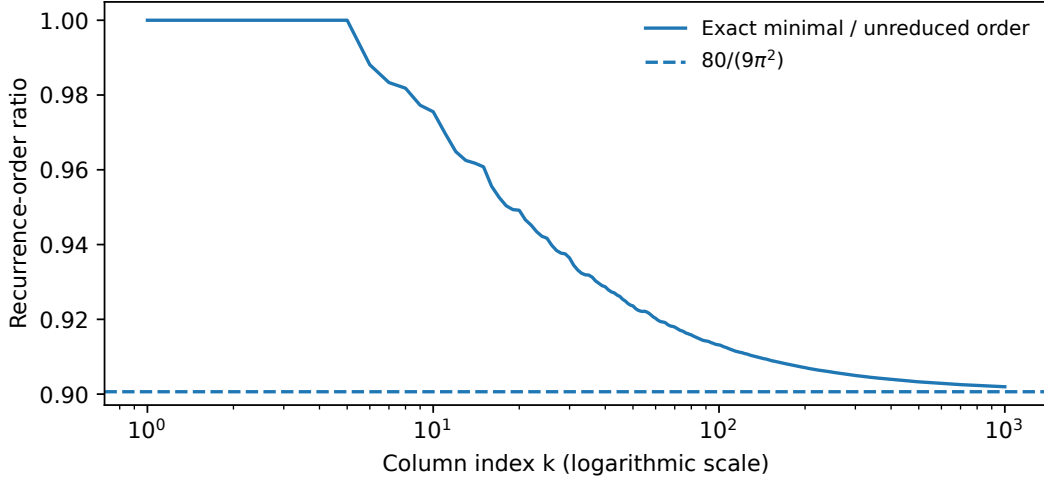


Figure 1: The recurrence-order ratio computed from the exact totient formula, not from numerical root finding. The dashed line is the proved limit.

6.1 The product denominator is not generally reduced

The first collision is between D_1 and D_7 :

$$D_1 = 1 - x, \quad D_7 = (x - 1)(x^2 - x - 1)(x^4 + x^3 - 4x^2 - 4x + 1).$$

Consequently,

$$R_6 = \mathcal{P}_6 / (1 - x), \quad r_6 = 83 < 84,$$

and $R_7 = \mathcal{P}_7 / (1 - x)^2$, $r_7 = 118 < 120$. The latter cancellation was already noticed in [5, Remark 5.7]. Theorem 5.3 gives the complete all-column replacement for a case-by-case cancellation check.

6.2 A numerator-degree index shift

A005586 has value $a(u) = u(u + 4)(u + 5)/6$ at index u [3]. The unreduced numerator degree in (22) is therefore

$$\deg G_k = A005586(k - 1) \quad (k \geq 1),$$

not $A005586(k)$ as the literal zero-based wording of Conjecture 4(ii) in A205497 suggests. For example, $G_1 = 1$ has degree 0, not 5. The explicit polynomial degree formula itself is already present in [5]; this is an indexing correction to the entry, not a new degree theorem.

6.3 The sign of the last coefficient in column three

The numerator of C_3 is

$$\begin{aligned} N_3(x) = & 1 + x - 6x^2 - 15x^3 + 21x^4 + 35x^5 - 13x^6 - 51x^7 \\ & + 3x^8 + 21x^9 + 5x^{10} + x^{11} - 5x^{12} - x^{13} + x^{14}. \end{aligned} \quad (31)$$

The final sign is *positive*. The formula in A205497's Conjecture 5.4 has a negative sign at x^{14} [1]; the positive sign is already printed in [5, Example 5.11]. The discrepancy first changes the coefficient of x^{14} in the generating function, since the common denominator has constant term 1:

$$[x^{14}]C_3(x) = M_{14,3} = z(19, 3) = 546629054.$$

The negative-sign version gives 546629052. The former also agrees with the independently listed A205493 term [4]. The supplied code verifies the exact difference $2x^{14}$ between the two numerators, rather than silently replacing the source data.

6.4 The known limit as a corollary of a stronger expansion

The Chebyshev expression in the old column-ratio conjecture equals

$$\rho_{k+1} = \frac{1}{2 \sin(\pi/(4k+6))} = U_k \left(\cos \frac{\pi}{2k+3} \right). \quad (32)$$

Indeed, $U_k(\cos \beta) = \sin((k+1)\beta)/\sin \beta$, and $(k+1)\beta = \pi/2 - \beta/2$ for $\beta = \pi/(2k+3)$. The positive coefficient α_{k+1} proves the corresponding ratio limit. The next section supplies error estimates and secondary terms rather than reclaiming this established limit.

7 Uniform spectral estimates

7.1 Bounds independent of matrix size

The key simplification is that the leading eigenvalue dominates all other ones by a factor bounded away from 1, uniformly in the alphabet size.

Lemma 7.1 (Uniform order-polynomial estimate). *For every $m, n \geq 1$,*

$$\left| \frac{\Omega_n(m)}{\alpha_m \rho_m^n} - 1 \right| \leq 4 \cdot 2^{-n}. \quad (33)$$

Also

$$\frac{3}{\pi} \leq \alpha_m \leq \frac{4}{3}, \quad 0 < \Omega_n(m) \leq \frac{\pi}{2} \rho_m^n. \quad (34)$$

Proof. For $m \geq 2$, the ratio of the second-largest eigenvalue in absolute value to ρ_m is

$$\frac{\sin t_m}{\sin 3t_m} = \frac{1}{3 - 4 \sin^2 t_m} < \frac{1}{2},$$

since $t_m \leq \pi/10 < \pi/6$. The orthogonal spectral decomposition and $\|\mathbf{1}\|^2 = m$ give

$$|\Omega_n(m) - \alpha_m \rho_m^n| \leq m(\rho_m/2)^{n-1}.$$

Because $m/\rho_m = 2m \sin t_m \leq 2mt_m < \pi/2$, division by $\alpha_m \rho_m^n$ gives the desired bound once the amplitude bounds are established. Now

$$\alpha_m = \frac{4}{\pi} \frac{t_m \cos^2 t_m}{\sin t_m}, \quad 0 < t_m \leq \pi/6.$$

The inequalities $\sin t \leq t$, $\cos^2 t \geq 3/4$, and $\sin t \geq 3t/\pi$ on this interval give $3/\pi \leq \alpha_m \leq 4/3$. Thus $m/(\alpha_m \rho_m) \leq \pi^2/6$ and the spectral relative error is at most $(\pi^2/3)2^{-n} < 4 \cdot 2^{-n}$. The norm bound $|\mathbf{1}^T H_m^{n-1} \mathbf{1}| \leq m \rho_m^{n-1}$ gives the final upper bound. When $m = 1$, the approximation is exact and all inequalities hold. $\square$

7.2 Concavity of the logarithmic growth rate

Extend the formulas for ρ_m and α_m to real $u \geq 1$. Write $y = u + 1/2$, $t = \pi/(4y)$, and

$$f(u) = \log \rho_u, \quad g(u) = \log \alpha_u, \quad \delta_u = f'(u) = \frac{t \cot t}{y}.$$

Lemma 7.2 (Derivative bounds). *For $u \geq 1$,*

$$\begin{aligned} f'(u) &> 0, & -\frac{2}{(u + 1/2)^2} &\leq f''(u) \leq -\frac{1}{2(u + 1/2)^2}, \\ |g''(u)| &\leq \frac{7}{(u + 1/2)^2}. \end{aligned} \tag{35}$$

In particular, for integers $1 \leq m \leq M$,

$$\rho_m / \rho_M \leq e^{-(M-m)\delta_M}. \tag{36}$$

As $u \rightarrow \infty$,

$$\delta_u = \frac{1}{u + 1/2} + O(u^{-3}). \tag{37}$$

Proof. Direct differentiation gives

$$y^2 f''(u) = t^2 \csc^2 t - 2t \cot t.$$

On $0 < t \leq \pi/6$, one has $t/\sin t \leq \pi/3$ and $t \cot t \geq \cos t \geq \sqrt{3}/2$. Therefore $y^2 f'' \leq \pi^2/9 - \sqrt{3} < -1/2$. The lower bound $y^2 f'' \geq -2$ follows from $t \cot t \leq 1$ and $t^2 \csc^2 t \geq 0$. For the amplitude, write

$$g(u) = \log(4/\pi) + h(t), \quad h(t) = \log(t/\sin t) + 2 \log \cos t.$$

A second differentiation gives

$$y^2 g''(u) = 1 - 2t \cot t - 4t \tan t + t^2 \csc^2 t - 2t^2 \sec^2 t.$$

Its absolute value is at most

$$1 + 2 + \frac{2\pi}{3\sqrt{3}} + \frac{\pi^2}{9} + \frac{2\pi^2}{27} < 7.$$

Since f' is decreasing, integrating it from m to M yields (36). Finally $t \cot t = 1 + O(t^2)$ proves (37). $\square$

7.3 An explicit relative-error bound

Theorem 7.3 (Uniform coefficient approximation). *For $n \geq 1$, $k \geq 0$, $M = k + 1$, and $q = e^{-n\delta_M}$,*

$$\left| \frac{z(n, k)}{\alpha_M \rho_M^n} - 1 \right| \leq 4 \cdot 2^{-n} + 2((1 + q)^{n+1} - 1). \tag{38}$$

If $(n + 1)q \leq 1$, the right side is at most

$$E(n, M) := 4 \cdot 2^{-n} + 4(n + 1)e^{-n\delta_M}. \tag{39}$$

For every fixed $0 < c < 1$, this error tends to zero uniformly for $M \leq cn/\log n$.

Proof. Separate the $j = 0$ term in (8). Its relative error is bounded by theorem 7.1. For $j \geq 1$, the upper bound in (34), the lower bound on α_M , and (36) give

$$\frac{\binom{n+1}{j} \Omega_n(M-j)}{\alpha_M \rho_M^n} \leq \frac{\pi^2}{6} \binom{n+1}{j} q^j < 2 \binom{n+1}{j} q^j.$$

Sum absolute values, extending to the full binomial sum. If $(n+1)q \leq 1$, use $(1+q)^{n+1} - 1 \leq e^{(n+1)q} - 1 \leq 2(n+1)q$.

For uniformity, δ_u decreases with u . At the largest permitted size $L = cn/\log n$,

$$n\delta_L = \frac{\log n}{c} + o(1).$$

Consequently $(n+1)e^{-n\delta_M} \leq n^{1-1/c+o(1)} \rightarrow 0$ uniformly over $M \leq L$. This also guarantees the hypothesis for the simpler bound once n is sufficiently large. $\square$

This proves part (i) of theorem 1.2. It also shows why the fixed column limit by itself is not enough for shape questions: the growing binomial coefficients in (8) must be controlled uniformly.

7.4 Two spectral terms for fixed columns

Proposition 7.4. *For each fixed $k \geq 2$,*

$$z(n, k) = \alpha_{k+1} \rho_{k+1}^n - (n+1) \alpha_k \rho_k^n + O_k(n^2 \rho_{k-1}^n). \quad (40)$$

For $k = 1$ the exact formula is $z(n, 1) = F_{n+2} - (n+1)$, where $F_0 = 0, F_1 = 1$ are the Fibonacci numbers.

Proof. The Perron terms at $m = k+1$ and $m = k$ in (17) give the two terms displayed. The Perron term at $m = k-1$ is of order $n^2 \rho_{k-1}^n$. Every Perron term at a smaller size has a strictly smaller exponential base, so its fixed polynomial factor can be absorbed in that bound. Among all non-Perron eigenvalues with $m \leq k+1$, the largest absolute value is $1/(2 \sin(3\pi/(4k+6)))$, and

$$\frac{1}{2 \sin(3\pi/(4k+6))} < \frac{1}{2 \sin(\pi/(4k-2))} = \rho_{k-1} \quad (k \geq 2).$$

Indeed the two angles lie in $(0, \pi/2)$ and the first is larger. All remaining spectral terms are therefore absorbed as well. For $k = 1$, use $\Omega_n(2) = F_{n+2}$ and $\Omega_n(1) = 1$ in (8). $\square$

8 The crossover at $k \asymp n/\log n$

The uniform bound identifies a transition scale. At that scale it is no longer correct simply to discard all terms with $j \geq 1$ in (8). Their combined effect has a closed limiting form.

Theorem 8.1 (Crossover law). *Let $k = k(n)$ be nonnegative integers such that*

$$s_n := \frac{n}{k+3/2} - \log n \rightarrow s \in \mathbb{R}. \quad (41)$$

Then

$$\frac{z(n, k)}{\alpha_{k+1} \rho_{k+1}^n} \rightarrow e^{-e^{-s}}. \quad (42)$$

Equivalently,

$$z(n, k) \sim \frac{4}{\pi} \left(\frac{2k+3}{\pi} \right)^n e^{-e^{-s}}. \quad (43)$$

Proof. Put $M = k + 1$. The hypothesis implies $M \sim n / \log n$, so $n/M^2 \rightarrow 0$, and (37) gives

$$n\delta_M - \log n = s_n + o(1) \rightarrow s.$$

For a fixed $j \geq 0$, Taylor expansion of $f = \log \rho$ on $[M - j, M]$ gives

$$n(f(M - j) - f(M)) = -nj\delta_M + O_j(n/M^2).$$

Also $\alpha_{M-j}/\alpha_M \rightarrow 1$, and theorem 7.1 is uniform in the changing size $M - j$. It follows that the absolute value of the j th normalized summand in (8) tends to $e^{-js}/j!$.

To justify summing these limits, define summands to be zero for $j \geq M$. For all $j \geq 0$, the same estimate used in theorem 7.3 gives

$$\frac{\binom{n+1}{j} \Omega_n(M - j)}{\alpha_M \rho_M^n} \leq 2 \frac{((n+1)e^{-n\delta_M})^j}{j!} \quad (0 \leq j < M).$$

The quantity $(n+1)e^{-n\delta_M}$ stays bounded. Thus a fixed summable majorant $2K^j/j!$ applies for all sufficiently large n . Dominated convergence for the series yields

$$\lim \frac{z(n, k)}{\alpha_M \rho_M^n} = \sum_{j=0}^{\infty} \frac{(-e^{-s})^j}{j!} = e^{-e^{-s}}.$$

Finally, $\alpha_M \rightarrow 4/\pi$ and

$$\rho_M = \frac{2M+1}{\pi} (1 + O(M^{-2})).$$

Since $n/M^2 \rightarrow 0$, taking the n th power leaves a relative factor tending to 1. This proves (43). $\square$

The function $e^{-e^{-s}}$ is the familiar Gumbel distribution function. Here it appears as a *deterministic relative coefficient correction*; no claim of a random-variable limit law is needed. At $s = 0$ the correction is e^{-1} , rather than 1. Thus the theorem describes precisely a regime where an unqualified leading Perron approximation would have a persistent relative error.

Corollary 8.2. *If $k/\sqrt{n} \rightarrow \infty$ and $k \log n/n \rightarrow 0$, then*

$$z(n, k) \sim \frac{4}{\pi} \left(\frac{2k+3}{\pi} \right)^n.$$

Proof. Use theorem 7.3, $\alpha_{k+1} \rightarrow 4/\pi$, and $n/(k+1)^2 \rightarrow 0$ in the expansion of ρ_{k+1} . $\square$

9 A growing part of the log-concavity conjecture

It is essential to distinguish uniformity in k from a separate eventual statement for every fixed k . The theorem below supplies the former.

Lemma 9.1 (Curvature of the leading approximation). *For every integer $m \geq 2$ and $n \geq 160$,*

$$\log \frac{(\alpha_m \rho_m^n)^2}{\alpha_{m-1} \rho_{m-1}^n \alpha_{m+1} \rho_{m+1}^n} \geq \frac{n}{4(m+3/2)^2}. \quad (44)$$

Proof. For any twice continuously differentiable function h ,

$$2h(m) - h(m-1) - h(m+1) = - \int_{-1}^1 (1-|v|)h''(m+v) dv.$$

Apply theorem 7.2 to $h = nf + g$. The result is bounded below by

$$\frac{n}{2(m+3/2)^2} - \frac{7}{(m-1/2)^2}.$$

For $m \geq 2$, $(m+3/2)/(m-1/2) \leq 7/3$, so the last expression is at least

$$\frac{n/2 - 343/9}{(m+3/2)^2} \geq \frac{n}{4(m+3/2)^2} \quad (n \geq 160).$$

□

Theorem 9.2 (Growing-edge strict log-concavity). *Fix $0 < c < 1/2$. There exists $n_0(c)$ such that every $n \geq n_0(c)$ and integer $k \geq 1$ with $k+2 \leq cn/\log n$ satisfy*

$$z(n, k)^2 > z(n, k-1)z(n, k+1).$$

For the same n , this also holds at $n-2-k$.

Proof. Let $L = cn/\log n$ and consider all three alphabet sizes $k, k+1, k+2 \leq L$ simultaneously. By theorem 7.3, their relative errors are at most

$$E_L = 4 \cdot 2^{-n} + 4(n+1)e^{-n\delta_L}.$$

For sufficiently large n , $E_L < 1/2$. Since $|\log(1+u)| \leq 2|u|$ for $|u| \leq 1/2$, the difference between the actual log-concavity logarithm and the leading one in (44) has absolute value at most $8E_L$. Consequently the actual logarithm is at least

$$\frac{n}{4(k+5/2)^2} - 8E_L. \tag{45}$$

The ratio of the second term to a constant multiple of the first is bounded by a constant times

$$\frac{(L+1/2)^2}{n} E_L = O(n 2^{-n}) + O\left(L^2 e^{-n\delta_L}\right) = n^{2-1/c+o(1)} O((\log n)^{-2}) + o(1).$$

This tends to zero because $c < 1/2$. Thus (45) is positive uniformly over all allowed k . The reflected statement follows from (6). □

Remark 9.3 (What is and is not resolved). The theorem proves infinitely many inequalities in a region whose width increases like $n/\log n$. It is strictly stronger than eventual log-concavity at each fixed distance from an edge. However, the width is still $o(n)$; the theorem does not reach a fixed positive fraction of a row or its center. It therefore does not settle Petersen–Zhuang Conjecture 6.1(b) in full. It does not prove real-rootedness, simple roots, or an interlacing/Sturm property either.

Corollary 9.4 (Every fixed edge index). *For each fixed integer $k \geq 1$,*

$$\frac{z(n, k)^2}{z(n, k-1)z(n, k+1)} \sim \frac{\alpha_{k+1}^2}{\alpha_k \alpha_{k+2}} \left(\frac{\rho_{k+1}^2}{\rho_k \rho_{k+2}} \right)^n,$$

and this quotient tends to infinity.

Proof. Apply the fixed-index cases of theorem 7.3. Strict concavity of $\log \rho_u$ implies $\rho_{k+1}^2 > \rho_k \rho_{k+2}$. □

10 Reproducible exact checks and numerical diagnostics

The accompanying scripts use no network access. Their roles are deliberately separated: integer and rational identities test algebraic implementation, while floating-point evaluations illustrate analytic results.

10.1 Exact checks

The default run of `verify.py` (shipped as `code/verify.py`) completed the following checks:

- (1) Direct symbolic determinants agree with (16) for $m = 1, \dots, 9$. Exact polynomial gcds agree with theorem 5.1 for all 820 unordered pairs $1 \leq m, n \leq 40$.
- (2) Direct enumeration of weak alternating words agrees with the matrix walk count for $m = 1, \dots, 4$ and $n = 0, \dots, 6$. Direct enumeration of alternating permutations and their big returns agrees with the extracted triangle for $n = 0, \dots, 9$.
- (3) Exact numerator and reduced-denominator certificates are produced for columns $k = 0, \dots, 12$. Each pair is checked for coprimality, the predicted degrees, and at least $2r_k + 26$ independently computed coefficients.
- (4) All rows through $n = 100$ were checked for symmetry, positivity, and log-concavity. These finite row checks are evidence only, not a proof of the full conjecture. The specified OEIS numerator discrepancy is also tested.

No check failed in the final run. The initial comparison against the printed column-three numerator detected the sign discrepancy explained in section 6; the final test records that discrepancy explicitly.

The certificates are in `data/rational_certificates.json`, with coefficients in ascending powers of x . The file `data/minimal_orders.txt` gives r_k for $0 \leq k \leq 1000$. It is candidate sequence data, not a claim that a new OEIS identifier has been assigned. The full machine-readable result is `data/verification_report.json`.

10.2 How the rational certificates are constructed

For $F_m = E_m/D_m$, one may avoid large symbolic expression simplifications. For $j \geq 1$,

$$\sum_{n \geq 0} \binom{n+1}{j} \Omega_n(m) x^n = \frac{x^{j-1}}{j!} \frac{d^j}{dx^j} (x F_m(x)). \quad (46)$$

Start with numerator $S = x E_m$ and denominator D_m^p , $p = 1$. One differentiation updates

$$S \mapsto S' D_m - p S D'_m, \quad p \mapsto p + 1.$$

After j steps, multiply by $x^{j-1}/j!$. For column k , multiply each resulting numerator by the exact polynomial quotient R_k/D_m^{k+2-m} , sum with alternating signs, and divide by x^{k+2} . This produces N_k using only exact polynomial arithmetic. The $j = 0$ and $k = 0$ cases are handled separately as above.

For coefficient verification, a suffix-sum multiplication by H_m takes $O(m)$ integer additions. Thus all $\Omega_n(m)$ with $n \leq N$ and $m \leq M$ can be generated in $O(NM^2)$ additions. This is an arithmetic-operation count; large integer bit lengths must still be included in a bit-complexity analysis. A recurrence built from R_k is useful for a long fixed column, but the approximate 9.94% degree reduction alone does not establish the best wall-clock algorithm.

10.3 Uniform-approximation diagnostics

Table 2 uses exact integer $z(n, k)$ and high-precision evaluation of the formulas. The error-bound column evaluates the rigorous expression (38), but its decimal evaluation is not interval-certified.

Table 2: Relative Perron errors and log-concavity logarithms. The last column is $\log(z(n, k)^2 / (z(n, k-1)z(n, k+1)))$.

n	k	Actual relative error	Bound expression	Logarithm
50	2	3.622e-06	8.108e-05	4.044101
100	5	6.153e-06	4.534e-05	2.361166
200	10	2.598e-06	1.156e-05	1.511073
500	20	2.303e-08	8.044e-08	1.081403
1000	40	2.562e-08	6.883e-08	0.580597
2000	80	3.786e-08	8.812e-08	0.301098

10.4 Crossover diagnostics

Table 3 illustrates theorem 8.1. Let $W = z(n, k) / (\alpha_{k+1} \rho_{k+1}^n)$. For these larger sizes we compute its finite alternating Perron sum, rather than construct the enormous exact integer. The omitted tail and the non-Perron contribution have explicit analytic bounds; floating-point roundoff is a separate issue, and the table is only a numerical diagnostic.

Specifically, put $K = (n+1)e^{-n\delta_M}$ and retain $0 \leq j < 100$. The absolute analytic remainder is at most

$$2e^K \frac{K^{100}}{100!} + 8 \cdot 2^{-n} e^K.$$

The first term bounds the factorial tail; the second follows from the uniform spectral bound. These are not bounds on the error of replacing W by the *limiting* crossover prediction.

Table 3: Approach to the crossover correction. Here $s_n = n/(k+3/2) - \log n$; W is evaluated by the truncated Perron sum.

n	k	s_n	W	$e^{-e^{-s_n}}$
1000	168	-1.00805	0.05858841	0.06455401
1000	143	0.01266	0.37213775	0.37253665
1000	125	0.99738	0.69634269	0.69153368
10000	1216	-0.99679	0.06547651	0.06656577
10000	1084	0.00200	0.36856637	0.36861673
10000	978	0.99895	0.69274428	0.69193318
100000	9511	-1.00044	0.06573410	0.06590880
100000	8684	0.00052	0.36806410	0.36806944
100000	7990	1.00037	0.69241787	0.69229481

11 A formalization route for ProveIt

A practical formalization should keep three boundaries separate.

Finite combinatorics and algebra. Define weak alternating words, the matrix H_m , and the suffix-sum evaluator. Prove their counting equivalence and then the descent/order-polynomial identity. A finite polynomial checker can verify each concrete pair (N_k, R_k) using rational arithmetic. This finite layer does not require trigonometric analysis.

Uniform algebraic structure. Formalize the sine eigenvectors or an equivalent Chebyshev description, including the nonzero projection of $\mathbf{1}$ onto every eigenspace. The noncancellation argument is then a general lemma about finite sums of exponential-polynomial sequences with distinct multiplier degrees. The gcd formula and first-occurrence factor theorem provide the structural content that finite examples cannot certify.

Analysis. The derivative inequalities, explicit error estimate, dominated convergence for the crossover, and finite-difference curvature bound can be developed independently of a full theory of roots of zig-zag Eulerian polynomials. Their statements should retain quantifiers over growing index ranges; replacing a uniform theorem by a pointwise limit would lose the essential content of theorem 9.2.

The included files are not untested Lean code presented as a formal proof. A future formalization should report exactly which of these boundaries has been crossed, and which imported results or analytic facts remain assumptions.

12 Further research questions

The results suggest several concrete projects. They are deliberately stated as questions or proposed programs, not as established consequences.

12.1 Full-row log-concavity and its central regime

Can the growing-edge theorem be joined to a local limit or saddle-point analysis valid when k/n stays in a compact subinterval of $(0, 1)$? A pointwise central approximation is insufficient: the error must be smaller than the second finite difference of the logarithm of the main term. An overlap theorem would be especially valuable because it could establish eventual full-row log-concavity, after which a finite certificate might handle the remaining rows. Neither step has been supplied here.

12.2 Beyond the constant $1/2$ in the edge theorem

The restriction $c < 1/2$ comes from bounding the three logarithmic errors separately. It is a limitation of this estimate, not an identified phase transition in log-concavity. Can one control the *second difference of the error* directly? The crossover formula suggests that the individual errors may vary smoothly enough for much better cancellation in that second difference. An explicit theorem for $c \geq 1/2$ is a well-defined next target.

12.3 A quantitative crossover expansion

Prove a rate and additional terms in theorem 8.1, uniformly for s_n in a fixed compact interval. The fixed- j expansion involves $nj^2/(k + 3/2)^2$, which is of order $(\log n)^2/n$ at crossover. One can expect the next terms to arise from exponential sums of j , j^2 , and higher powers against $(-e^{-s})^j/j!$. A useful result would give an explicit uniform remainder, not just fitted coefficients. The finite-index rounding of k must be retained through s_n .

12.4 A direct combinatorial meaning of the crossover correction

The exponential series in the proof resembles a sieve over rare defects. Is there a combinatorial construction whose limiting exclusion probability is $e^{-e^{-s}}$ and whose exact count is $z(n, k)$ after normalization? Such a construction could explain the correction without relying solely on signed coefficient extraction, and might lead to estimates deeper inside the triangle.

12.5 Sharper recurrence-order asymptotics

The current remainder $O(k^2 \log k)$ is enough to identify the leading constant but does not resolve a secondary term. The exact totient sum reduces the problem to weighted odd-totient summatory estimates. Can one describe the secondary behavior, including any arithmetic fluctuations, with a useful explicit error? A correct answer must not infer a polynomial second term from a short table.

12.6 Minimal realizations and fast exact generation

The degree r_k determines the minimum linear state dimension over a field, but a dense recurrence need not be the fastest realization over the integers. Can the factor decomposition in (26) be exploited by modular polynomial arithmetic, block recurrences, or rational reconstruction? The target is a proved bit-complexity bound and a reproducible comparison against suffix-sum dynamic programming, with output size accounted for.

12.7 Other periodic fence orientations

Replace the alternating orientation by a fixed periodic pattern of up and down comparisons. The order-polynomial transfer matrix becomes a product of upper and lower summation matrices. Which patterns retain an explicit spectrum, and which permit a first-occurrence classification strong enough to determine every fixed-descent minimal recurrence? The noncancellation principle from section 4 survives whenever the spectral weights and the polynomial multipliers can be controlled; the difficult part is identifying and comparing the spectra.

12.8 Real-rootedness and refined interlacing

The full real-rootedness conjecture remains separate from the present argument. A direct stability-preserving recurrence, an interlacing refinement, or a sign-regular matrix model would be stronger than gamma-positivity alone. The refined recurrences of [6] provide a concrete starting point, but an assertion that they preserve real roots requires a proof of every needed interlacing hypothesis. The current spectral formulas concern the index n in fixed columns and do not automatically control zeros in the descent variable t .

13 Conclusion

The fixed-column theory of A205497 has an exact algebraic refinement: its minimum recurrence is controlled by the *earliest appearance* of each reciprocal eigenvalue, not by multiplying all available denominators. That observation yields a complete collision classification and an explicit cubic order constant.

The same spectral representation has a second use. Uniform control of alphabet size makes it possible to pass from fixed-column asymptotics to a crossover law and to strict log-concavity

in a growing edge region. The remaining central and real-rootedness questions are clearly delimited. The package supplies ordinary proofs, exact reproducible computations, and specific next problems; it does not convert limited checks or a literature search into a claim of a fully formalized or globally priority-certified breakthrough.

A A finite, explicit log-concavity certificate from the bounds

The existence statement in theorem 9.2 can be used without guessing $n_0(c)$. For given integers $n \geq 160$ and $k \geq 1$, put $L = k + 2$ and $q_L = e^{-n\delta_L}$. If

$$(n+1)q_L \leq 1, \quad E(n, L) < \frac{1}{2}, \quad 8E(n, L) < \frac{n}{4(k+5/2)^2}, \quad (47)$$

then $z(n, k)^2 > z(n, k-1)z(n, k+1)$. Indeed, the errors at all three sizes are bounded by $E(n, L)$ because δ_u decreases, and (45) is positive. These inequalities involve elementary real functions only; an interval implementation could make a formally checkable numerical certificate for specific ranges. The decimal diagnostics supplied here do not implement that interval certification.

B Package contents and reproduction

The main source is `zigzag_spectral_research.tex`; its bibliography is embedded below, so no external bibliography database is required.

```
python verify.py --out-dir data
python illustrate.py
latexmk -pdf -interaction=nonstopmode -halt-on-error \
    zigzag_spectral_research.tex
```

`build.sh` executes these steps. Python dependencies are SymPy, mpmath, and Matplotlib; the final recorded run used Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0. The exact checks can be rerun separately from `illustrate.py`. The generated plot, small typeset tables, machine-readable certificates, triangle values, and numerical diagnostics are in `data/`. No font files or external copyrighted articles are included.

[Write note, batch 7302, 1 October 2026.] The paragraph above describes the delivered package. In ProveIt the source is `article.tex`, the scripts and both build wrappers are in `code/`, and the recorded outputs and `requirements.txt` are in `data/`. The delivered commands, and `build.sh` and `build.ps1`, rewrite the recorded files in `data/`, and after the move `code/verify.py` by default writes to `code/data/`; the report's `README.md` gives commands that run on a copy. The figure `data/order_ratio.pdf` was regenerated from the unmodified `illustrate.py` with TrueType instead of Type 3 fonts; its content is unchanged (see the `README`).

C Provenance

This report is one manuscript, number 36 of batch 7302 of ProveIt's incoming-reports intake: the archive `zigzag_oeis_research.zip` (main file `zigzag_spectral_research.tex`, 22-page PDF), dated 1 October 2026, which arrived in commit `f8c3a392a` and was placed by commit `6e193dd4f`. It cites no repository revision: its only repository source is the ProveIt `README`,

inspected on 1 October 2026. The text is as delivered except for the label prefix **zz:**, the citation of A050446 (listed but uncited in the delivered bibliography) in Section 2, the shipped path of the verification script in Section 10, this appendix, the repository bibliography entry, and the dated write notes in Section 1 and Appendix B.

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